

B.Sc. Semester-IV (Honours) Examination, 2023-24

COMPUTER SCIENCE

Course ID: 41513

Course Code: SH/CSC/403/C-10

Course Title: Database Management Systems

[SYLLABUS 2022-23]

Time: 1 Hour 15 minutes.

Full Marks: 25

The figures in the margin indicate full marks.

Answer all the questions

UNIT-I

1. Answer *any five* of the following questions: (5×1 = 5)
- a) What is Primary Key?
 - b) Define SELECT operation in Relational algebra.
 - c) Define BCNF.
 - d) Define 3rd normal form
 - e) Define transaction
 - f) List the DBMS applications
 - g) What is DML?
 - h) Define attribute.

UNIT-II

2. Answer *any two* of the following questions: (5×2=10)
- a) Explain in brief ANSI/SPARC three level architecture in DBMS
 - b) Explain about Integrity constraints over relations in detail.
 - c) What are the conditions required for a relation to be in 2NF and 3NF?
 - d) Explain the ACID properties of transaction.

[Turn Over]

UNIT-III

3. Answer *any one* of the following questions: (1×10 = 10)

a) Explain deletion and search operation in B+ trees,

b) Consider the following relational schemas:

student(sid, sname, saddress, DOB, cid)

course(cid, cname, dept_id, duration_yr)

department(dept_id, dept_name)

Create the tables and write query to perform the following tasks:

- (i) List the detail of all students whose birth date in on or before 1st January, 1998,
- (ii) Retrieve the name of the students who are studying 'B.Sc. Computer Science'.
- (iii) Change the name of the student to 'Indra' whose ID is 201.
- (iv) Retrieve only the names of the students who has taken admission in courses having duration less than 3 years.

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[SYLLABUS 2017-18]

Time: 1 Hour 15 minutes.

Full Marks: 25

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Answer all the questions

UNIT - I

1. Answer *any five* of the following questions: (5×1 = 5)
- a) What is Super Key?
 - b) What is the purpose of GROUP BY clause in SQL?
 - c) What is the purpose ORDER BY clause in SQL?
 - d) What do you mean by functional dependency in database?
 - e) What is the role of join operation in relational algebra?
 - f) What is domain integrity?
 - g) Define UNION operation.
 - h) Define Normalization.

UNIT - II

2. Answer *any two* of the following questions: (5×2=10)
- a) Explain in brief ANSI/SPARC three level architecture in DBMS.
 - b) Explain the different types of constraints in DBMS with example.
 - c) State and explain various features of E-R models.
 - d) Briefly explain the role of a DBA.

[Turn Over]

UNIT - III

3. Answer *any one* of the following questions: (1×10 = 10)

a) Define BCNF. How does it differ from 3NF? Why is it considered a stronger form of 3NF?

b) Differentiate between the following: 2×5 = 10

- (i) Stored attributes and derived attributes
 - (ii) Strong entity and weak entity.
 - (iii) DDL and DML
 - (iv) Equijoin and natural join
 - (v) SELECT and PROJECT operation in relational algebra.
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BANKURA SAMMILANI COLLEGE

Internal Assessment Examination-2024

Semester-4th (C-10)

Paper-DBMS

Full Marks-10

Time-30 min

Answer any five questions:

5*2=10

1. Write down two advantages of using DBMS.
2. What is Primary key?
3. What is data independence?
4. What is weak entity and what is strong Entity?
5. What are the different kinds of attribute available in ER model?
6. What are the DDL and DML commands in SQL?
7. Write down a SQL query to create STUDENT table with suitable attribute.
8. Write down the levels that three schema architecture contains.

B.Sc. Semester -IV Examination, 2023-24**Subject : Computer Science****Course ID :41523****Course Code: SH/CSC/403/C-10****Course Title: Database Management Systems****(Syllabus:2017 – 18 & 2022 - 23)****Time: 2 Hours****Full Marks :15***The figures in the right-hand margin indicate marks.**Candidates are required to give answers in their own words as far as practicable***[Problem = 10 marks, LNB + Viva Voce = 5 marks]****Create and use the following database schema to answer the given queries.**

EMPLOYEE Schema				
Field	Type	NULL	KEY	DEFAULT
Eno	Char(3)	NO	PRI	NIL
Ename	Varchar(50)	NO		NIL
Job_type	Varchar(50)	NO		NIL
Manager	Char(3)	YES	FK	NIL
Hire_date	Date	NO		NIL
Dno	Integer	YES	FK	NIL
Commission	Decimal (10,2)	YES		NIL
Salary	Decimal (7,2)	NO		NIL

[Turn Over]

DEPARTMENT Schema					9. C
Field	Type	NULL	KEY	DEFAULT	dep
Dno	Integer(3)	NO	PRI	NULL	10.
Dname	Varchar(50)	YES		NUL	emp
Location	Varchar(50)	YES		NEW DELHI	

Answer any one of the following questions:

1x10 = 10

1. Query to display Employee Name and Salary for all employees whose salary is not in the range of \$1500 and \$2850.

2. Query to display Employee Name and department No. of all the employees in Dept 10 and Dept 30 in the alphabetical order by name.

3. Query to display Name of all employees either have two R's or have two A's in their name and are either in Dept No = 30 or their Manager's Employee No = 7788.

4. Query to display Name, Salary and Commission for all employees whose Commission amount is 14 greater than their Salary increased by 5%.

5. Query to display Name, Job, Department No. And Department Name for all the employees working at the Dallas location.

6. Query to display Name, Dept No. And salary of any employee whose department No. and salary matches both the department no. And the salary of any employee who earns a commission.

7. Query to display Employee Number and Name for all employees who work in a department with any employee whose name contains a T.

8. Query to display the names and salaries of all employees who report to King.

9. Query to display the department no, name and job for all employees in the Sales department.

10. Query to display the highest, Lowest, Sum and Average Salaries of all the employees.